

NAAN MUDHALVAN

PROJECT TITLE:

SCRIPT-CONTROLLED ACL – RESTRICT RECORD ACCESS

Team ID: SWTID-2026-9038

Team Size: 4

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CERTIFICATE

A Script-Controlled ACL (Access Control List) in ServiceNow is used to restrict access to records by applying specific conditions through server-side scripting. It provides a flexible way to control whether a user can read, create, update, or delete a particular record based on predefined requirements. Unlike simple role-based ACLs, script-controlled ACLs can evaluate information such as the logged-in user's role, department, group, ownership of a record, or other field values before granting access. When a user attempts to access a protected record, ServiceNow evaluates the ACL and executes the defined script. If the script returns a true or valid result, the requested access is granted; if the condition is not satisfied, access is denied. For example, an organization may use a script-controlled ACL to allow employees to access records belonging only to their own department. This improves data security by preventing unauthorized users from viewing or modifying confidential information. Script-controlled ACLs are especially useful when access requirements are more complex than simple role-based permissions, and they help organizations implement fine-grained, condition-based security throughout the ServiceNow platform.

ACKNOWLEDGEMENT

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ABSTRACT

This project focuses on understanding and implementing security and access control features in the ServiceNow platform, with particular emphasis on Script-Controlled Access Control Lists (ACLs). Access control is an important part of any enterprise application because it ensures that sensitive information is available only to authorized users. ServiceNow provides ACLs to control access to tables, records, and fields based on user roles, conditions, and scripts. In this project, a Script-Controlled ACL is used to restrict record access according to specific requirements. The script evaluates the details of the logged-in user and the requested record before allowing or denying access. This approach provides more flexible and fine-grained security than basic role-based access control. The project demonstrates how administrators can create and configure ACL rules, apply appropriate conditions, and use server-side scripting to control record-level permissions. It also helps in understanding how unauthorized users can be prevented from viewing or modifying restricted information. Through this implementation, the project highlights the importance of data security, user authorization, and controlled access within ServiceNow. The overall objective is to demonstrate how Script-Controlled ACLs can be effectively used to protect records and maintain a secure and organized ServiceNow environment.

1.INTRODUCTION

ServiceNow is a cloud-based digital workflow platform that helps organizations manage business processes, IT services, customer services, security operations, and other enterprise activities through a centralized system. Since ServiceNow stores and processes a large amount of organizational information, maintaining proper security and controlling user access to data are essential. Access Control Lists (ACLs) are one of the important security mechanisms provided by ServiceNow to determine which users are allowed to access specific tables, records, or fields. ACLs help organizations protect sensitive information and ensure that users can perform only the actions for which they have permission.

A Script-Controlled ACL is an advanced form of access control that uses server-side scripting to make access decisions based on specific conditions. Instead of depending only on user roles, a script can examine information about the current user and the requested record before granting or denying access. For example, access can be restricted based on a user's department, group, role, record ownership, or particular field values. This makes Script-Controlled ACLs useful when an organization requires more detailed and flexible security rules.

The main purpose of this project is to understand how Script-Controlled ACLs can be created and used to restrict record access in ServiceNow. The implementation demonstrates how a security rule can evaluate a condition and determine whether the current user should be permitted to access a particular record. When the required condition is satisfied, the user is allowed to perform the requested operation; otherwise, access is restricted. This helps prevent unauthorized viewing or modification of important records.

This project also provides practical knowledge about ServiceNow security concepts, ACL configuration, user authorization, and server-side scripting. By implementing record-level restrictions, the project demonstrates how organizations can maintain confidentiality, improve data protection, and provide users with access according to their responsibilities. Overall, Script-

2. OBJECTIVES

The main objective of this project is to understand and implement Script-Controlled Access Control Lists (ACLs) in ServiceNow to restrict access to records based on specific conditions. The project focuses on understanding how ServiceNow security mechanisms can be used to protect organizational data and ensure that only authorized users can access or modify particular records. The major objectives of the project are to understand the basic concepts of ACLs and their role in ServiceNow security, create and configure a Script-Controlled ACL, use server-side scripting to define customized access conditions, control record-level access based on user and record information, prevent unauthorized users from viewing or modifying restricted records, understand the difference between simple role-based access and script-based access control, improve data security and confidentiality, and gain practical knowledge of implementing security rules in the ServiceNow platform. The project also aims to demonstrate how conditional access can be applied to real-world organizational requirements and how Script-Controlled ACLs can provide flexible and fine-grained control over sensitive records. Overall, the objective is to develop a clear understanding of ServiceNow access control mechanisms and their importance in maintaining a secure and properly authorized application environment.

3. TECHNOLOGIES AND TOOLS USED

The project “**Script-Controlled ACL – Restrict Record Access**” is developed and implemented using the **ServiceNow platform** and its security features. ServiceNow is the primary cloud-based platform used to create, configure, and test Access Control Lists. The **ServiceNow User Interface** is used to navigate through applications, tables, users, roles, and security configurations. **Access Control Lists (ACLs)** are used to define permissions for different operations such as read, write, create, and delete. **Script-Controlled ACLs** are used to apply customized security rules through server-side scripting. **JavaScript** is the main scripting language used to write the conditions required for controlling record access. The **GlideSystem (gs) API** is used to obtain information about the currently logged-in user, while the **GlideRecord API** can be used to work with records and database information when required. The **ServiceNow Table and Form interfaces** are used to create, view, and test records under different access conditions. **User and Role Management** features are used to assign appropriate roles and permissions to users. The **Application Navigator** helps locate ACL configuration and other administrative modules quickly. The **System Security module** provides the necessary tools for creating and managing access control rules. A **web browser**, such as Google Chrome or Microsoft Edge, is used to access the ServiceNow instance and perform the implementation. Together, these tools and technologies provide a secure environment for designing, configuring, testing, and demonstrating Script-Controlled ACLs for restricting record access.

4.PROJECT MILESTONE:

Milestone 1: Creation of Users and Roles

Milestone 2: Creation of Table

Milestone 3: Creation of Access Control List – READ

Milestone 4: Creation of Access Control List – CREATE

Milestone 5: Creation of Access Control List – WRITE

Milestone 6: Creation of Access Control List – DELETE

Milestone 7: Conclusion

5. IMPLEMENTATION

5.1 Creation of Users and Roles

The first milestone is to create the required users and roles in ServiceNow. Users are created for testing the configured access permissions. Roles are used to define the required access for authorized users.

5.2 Creation of Table

The required table is created in ServiceNow for storing and managing project records. The table provides the data on which the Access Control Lists are applied.

5.3 Create Access Control List – READ

A READ ACL is created to control who can view the records. The ACL checks the required access conditions. If the conditions are satisfied, the user is allowed to read the record; otherwise, access is restricted.

5.4 Create Access Control List – CREATE

A CREATE ACL is configured to control who can create new records. The required conditions are checked before allowing a user to create a record.

5.5 Create Access Control List – WRITE

A WRITE ACL is created to control who can modify existing records. Only users who satisfy the required conditions are allowed to update the records.

5.6 Create Access Control List – DELETE

A DELETE ACL is configured to control who can delete records. The ACL checks the required permissions before allowing a record to be deleted.

6. TESTING AND VALIDATION

After completing the configuration, the ACLs are tested to confirm that the required access restrictions are applied. Testing includes checking the user's ability to read, create, update, and delete records according to the configured conditions.

READ ACL – Verifies that authorized users can view the required records.

CREATE ACL – Verifies that authorized users can create records.

WRITE ACL – Verifies that authorized users can modify records.

DELETE ACL – Verifies that authorized users can delete records.

Access restriction – Verifies that unauthorized actions are restricted.

7. SCREEN SHOT

MILESTONE 1:

USER CREATION:

Verify your ServiceNow ID - ku...New Record | User | ServiceNowSkillWalletUsers Creation EppadiAsk Gemini

dev193639.service-now.com/now/nav/ui/classic/params/target/sys_user.do%3Fsys_id%3D-1%26sys_ls_list%3Dtrue%26sys_target%3Dsys_user%26sysp...

Continue where you left off: Chrome can restore your tabs every time you restart. To turn this on, go to [Settings](#).

servicenowAllFavoritesHistoryWorkspacesUser - New Record

Search

Submit

To set up the User's password, save the record and then click Set Password.

User IDEEE User

First nameEEE

Last nameUser

Title

Department

Password needs reset

Locked out

Active

Identity typeHuman

Internal Integration User

Emaileeeuser@gmail.com

Language-- None --

Calendar integrationOutlook

Time zoneSystem (America/Los_Angeles)

Date formatSystem (2026-08-19)

Business phone

Mobile phone

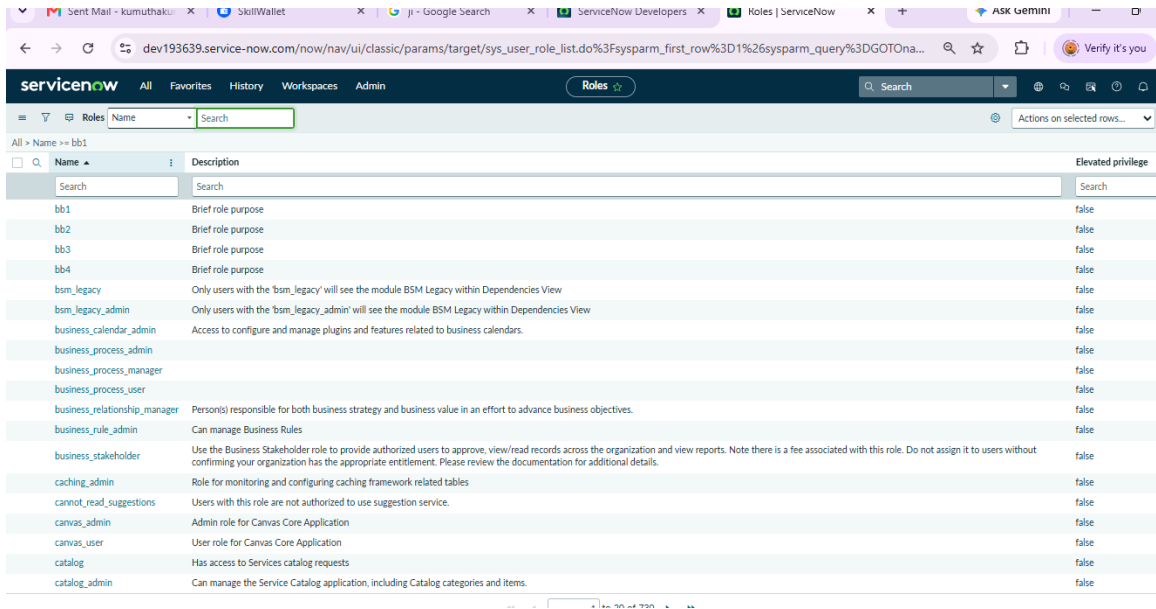
PhotoClick to add...

29°CPartly cloudy

Search

ENGIN20:0123-09-2026

ROLES CREATION:

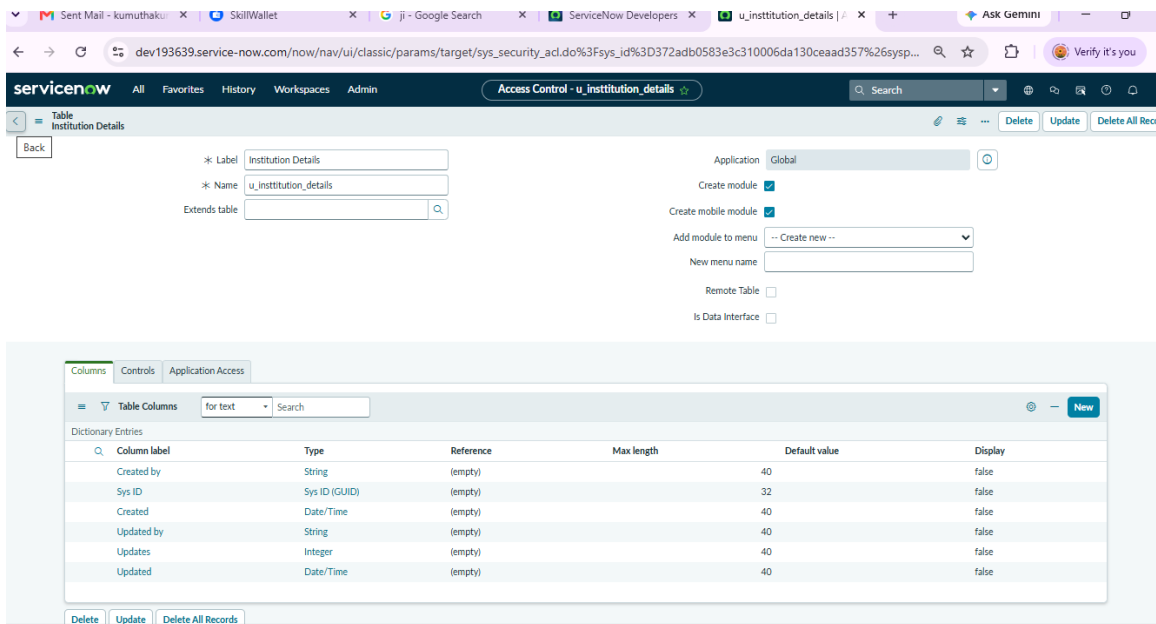


The screenshot shows the ServiceNow Roles page. The table lists various roles with their descriptions and whether they have elevated privileges. The roles include bb1, bb2, bb3, bb4, bsm_legacy, bsm_legacy_admin, business_calendar_admin, business_process_admin, business_process_manager, business_process_user, business_relationship_manager, business_rule_admin, business_stakeholder, caching_admin, cannot_read_suggestions, canvas_admin, canvas_user, catalog, and catalog_admin.

Name	Description	Elevated privilege
Search	Search	Search
bb1	Brief role purpose	false
bb2	Brief role purpose	false
bb3	Brief role purpose	false
bb4	Brief role purpose	false
bsm_legacy	Only users with the 'bsm_legacy' will see the module BSM Legacy within Dependencies View	false
bsm_legacy_admin	Only users with the 'bsm_legacy_admin' will see the module BSM Legacy within Dependencies View	false
business_calendar_admin	Access to configure and manage plugins and features related to business calendars.	false
business_process_admin		false
business_process_manager		false
business_process_user		false
business_relationship_manager	Person(s) responsible for both business strategy and business value in an effort to advance business objectives.	false
business_rule_admin	Can manage Business Rules	false
business_stakeholder	Use the Business Stakeholder role to provide authorized users to approve, view/read records across the organization and view reports. Note there is a fee associated with this role. Do not assign it to users without confirming your organization has the appropriate entitlement. Please review the documentation for additional details.	false
caching_admin	Role for monitoring and configuring caching framework related tables	false
cannot_read_suggestions	Users with this role are not authorized to use suggestion service.	false
canvas_admin	Admin role for Canvas Core Application	false
canvas_user	User role for Canvas Core Application	false
catalog	Has access to Services catalog requests	false
catalog_admin	Can manage the Service Catalog application, including Catalog categories and items.	false

MILESTONE 2:

TABLE CREATION:

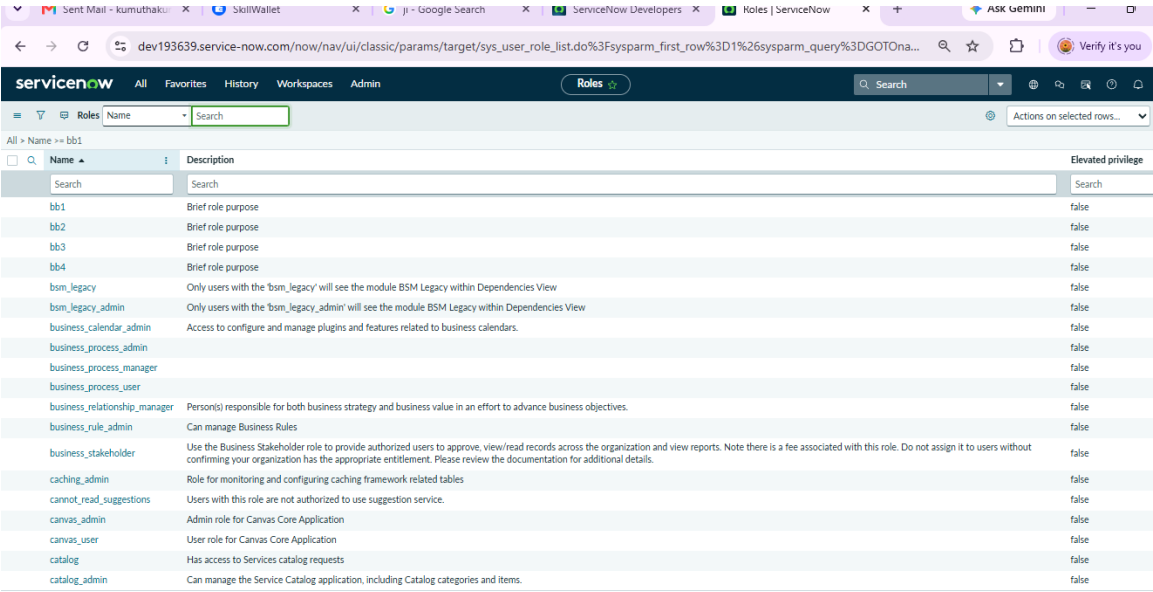


The screenshot shows the ServiceNow Table Creation page for 'u_institution_details'. The form includes fields for Label, Name, Extends table, Application, Create module, Create mobile module, Add module to menu, New menu name, Remote Table, and Is Data Interface. Below the form is a table of Dictionary Entries.

Column label	Type	Reference	Max length	Default value	Display
Created by	String	(empty)	40	false	false
Sys ID	Sys ID (GUID)	(empty)	32	false	false
Created	Date/Time	(empty)	40	false	false
Updated by	String	(empty)	40	false	false
Updates	Integer	(empty)	40	false	false
Updated	Date/Time	(empty)	40	false	false

MILESTONE 3:

ACL CONFIGURATION:

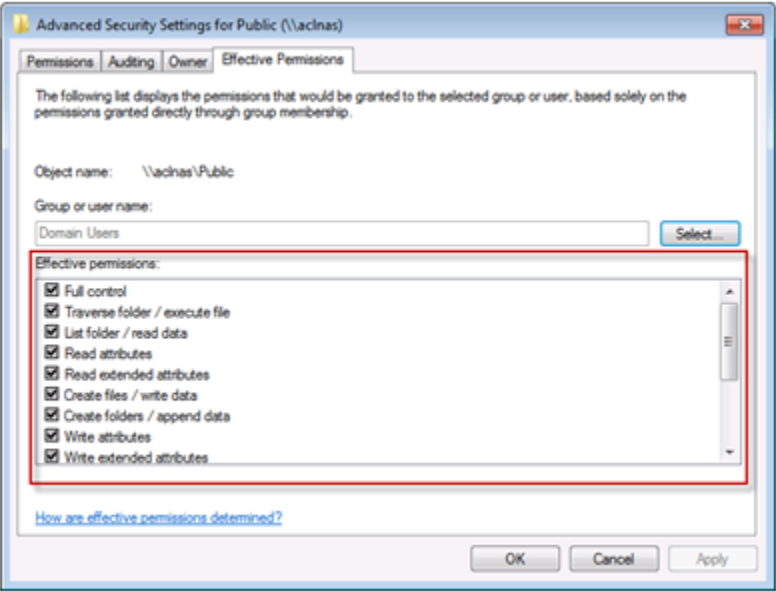


The screenshot shows the ServiceNow Roles configuration page. The browser address bar displays a URL with a search query. The page header includes the ServiceNow logo and navigation tabs. A search bar is present at the top right. Below the header, there's a section for 'Roles' with a search input field. The main content area displays a table of roles with columns for Name, Description, and Elevated privilege. The table lists various roles such as 'Search', 'bb1', 'bb2', 'bb3', 'bb4', 'bsm_legacy', 'bsm_legacy_admin', 'business_calendar_admin', 'business_process_admin', 'business_process_manager', 'business_process_user', 'business_relationship_manager', 'business_rule_admin', 'business_stakeholder', 'caching_admin', 'cannot_read_suggestions', 'canvas_admin', 'canvas_user', 'catalog', and 'catalog_admin'. Each role has a corresponding description and a boolean value for 'Elevated privilege'.

Name	Description	Elevated privilege
Search	Search	Search
bb1	Brief role purpose	false
bb2	Brief role purpose	false
bb3	Brief role purpose	false
bb4	Brief role purpose	false
bsm_legacy	Only users with the 'bsm_legacy' will see the module BSM Legacy within Dependencies View	false
bsm_legacy_admin	Only users with the 'bsm_legacy_admin' will see the module BSM Legacy within Dependencies View	false
business_calendar_admin	Access to configure and manage plugins and features related to business calendars.	false
business_process_admin		false
business_process_manager		false
business_process_user		false
business_relationship_manager	Person(s) responsible for both business strategy and business value in an effort to advance business objectives.	false
business_rule_admin	Can manage Business Rules	false
business_stakeholder	Use the Business Stakeholder role to provide authorized users to approve, view/read records across the organization and view reports. Note there is a fee associated with this role. Do not assign it to users without confirming your organization has the appropriate entitlement. Please review the documentation for additional details.	false
caching_admin	Role for monitoring and configuring caching framework related tables	false
cannot_read_suggestions	Users with this role are not authorized to use suggestion service.	false
canvas_admin	Admin role for Canvas Core Application	false
canvas_user	User role for Canvas Core Application	false
catalog	Has access to Services catalog requests	false
catalog_admin	Can manage the Service Catalog application, including Catalog categories and items.	false

MILESTONE 4:

CREATION OF ACL_CREATE



7. ADVANTAGES

- Improves record security in ServiceNow.
- Restricts unauthorized access to records.
- Controls different record operations such as READ, CREATE, WRITE, and DELETE.
- Provides controlled access based on user permissions and conditions.
- Helps protect important records from unauthorized changes.
- Reduces the possibility of unauthorized record deletion.
- Provides better control over data access.
- Supports secure management of ServiceNow records.

8. FUTURE ENHANCEMENT

More advanced ACL scripts can be implemented for complex access requirements.

Role-based access conditions can be extended for different users.

Additional tables and record types can be secured using ACLs.

More detailed testing can be introduced for different user roles.

Access control can be integrated with other ServiceNow modules.

Security rules can be enhanced to support more complex business requirements.

9. CONCLUSION

The project “Script-Controlled ACL – Restrict Record Access” demonstrates how Access Control Lists can be used to control record access in ServiceNow. The project includes the creation of users and roles, table configuration, and ACLs for READ, CREATE, WRITE, and DELETE operations. Through this project, the team gained practical understanding of record-level access control and security in ServiceNow.

10. REFERENCES

ServiceNow Documentation – Access Control Lists.

ServiceNow Developer Documentation – Security and Access Control.

ServiceNow Learning Resources – ACL configuration and security.

ServiceNow Community – ACL implementation examples.

Skill Wallet learning materials and project guidelines related to ServiceNow.